

```
from google.colab import drive
drive.mount('/content/drive')
```

 Mounted at /content/drive

```
!rm -rf /content/dataset
!mkdir /content/dataset
```

```
!unzip -o "/content/drive/MyDrive/H&H/parking/parking_slots.v2i.yolov8.zip" -d /content/dataset
!unzip -o "/content/drive/MyDrive/H&H/parking/parking_slots.v3i.yolov8.zip" -d /content/dataset
!unzip -o "/content/drive/MyDrive/H&H/parking/parking_slots.v4i.yolov8.zip" -d /content/dataset
!unzip -o "/content/drive/MyDrive/H&H/parking/parking_slots.v5i.yolov8.zip" -d /content/dataset
!unzip -o "/content/drive/MyDrive/H&H/parking/parking_slots.v6i.yolov8.zip" -d /content/dataset
```



```

inflating: /content/dataset/valid/labels/frame_00153_jpg.rf.6b5aedbb07965fbad9b217e6
inflating: /content/dataset/valid/labels/frame_00160_jpg.rf.c4f759982068d1a6e826c133
inflating: /content/dataset/valid/labels/frame_00167_jpg.rf.7b0fe69ce4c8fe11f8df89ed
inflating: /content/dataset/valid/labels/frame_00168_jpg.rf.c660652f954b2dc174915c4d
inflating: /content/dataset/valid/labels/frame_00170_jpg.rf.55844bc77639cd95b1a0b284
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inflating: /content/dataset/valid/labels/frame_00206_jpg.rf.520ece08b5f7f10331b33ac5
inflating: /content/dataset/valid/labels/frame_00226_jpg.rf.1214dc3c25e5acd301fde276
inflating: /content/dataset/valid/labels/frame_00251_jpg.rf.585e7be8672ec18c52e4eb08
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inflating: /content/dataset/valid/labels/frame_00256_jpg.rf.06fbd666dc9a2902864e1d8a
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inflating: /content/dataset/valid/labels/frame_00298_jpg.rf.9aaca73071e465bf0dd99b24
inflating: /content/dataset/valid/labels/frame_00305_jpg.rf.88b86130138db58fcb5f9288

```

```
!ls /content/dataset
```

```

➡ data.yaml  README.dataset.txt  README roboflow.txt  test  train  valid

```

```
!pip install ultralytics
```

```
➡ Collecting ultralytics
```

```
  Downloading ultralytics-8.3.189-py3-none-any.whl.metadata (37 kB)
```

```

Requirement already satisfied: numpy>=1.23.0 in /usr/local/lib/python3.12/dist-package
Requirement already satisfied: matplotlib>=3.3.0 in /usr/local/lib/python3.12/dist-pac
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Requirement already satisfied: polars in /usr/local/lib/python3.12/dist-packages (from
Collecting ultralytics-thop>=2.0.0 (from ultralytics)

```

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  Downloading ultralytics_thop-2.0.16-py3-none-any.whl.metadata (14 kB)
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Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-pack
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Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-pac
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Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-

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Requirement already satisfied: nvidia-cuda-cupti-cu12==12.6.80 in /usr/local/lib/pytho
Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in /usr/local/lib/python3.
Requirement already satisfied: nvidia-cublas-cu12==12.6.4.1 in /usr/local/lib/python3.
Requirement already satisfied: nvidia-cufft-cu12==11.3.0.4 in /usr/local/lib/python3.1
Requirement already satisfied: nvidia-curand-cu12==10.3.7.77 in /usr/local/lib/python3
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Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packa
Downloading ultralytics-8.3.189-py3-none-any.whl (1.1 MB)
```

1.1/1.1 MB 24.9 MB/s eta 0:00:00

Downloading ultralytics_thop-2.0.16-py3-none-any.whl (28 kB)

Installing collected packages: ultralytics-thop, ultralytics

Successfully installed ultralytics-8.3.189 ultralytics-thop-2.0.16

```
from ultralytics import YOLO
```

```
# Load YOLOv8 model (choose 'yolov8n.pt', 'yolov8s.pt', etc.)
```

```
model = YOLO("yolov8n.pt")
```

```
# Train
```

```
model.train(
    data="/content/dataset/data.yaml", # path to data.yaml
    epochs=50,                         # increase if dataset is big
    imgsz=640,                         # image size
    batch=16,                          # adjust if Colab RAM is low
    name="parking82_best"
)
```



```

0.80581,    0.80681,    0.80781,    0.80881,    0.80981,    0.81081,
0.81181,    0.81281,    0.81381,    0.81481,    0.81582,
          0.81682,    0.81782,    0.81882,    0.81982,    0.82082,    0.82182,
0.82282,    0.82382,    0.82482,    0.82583,    0.82683,    0.82783,
0.82883,    0.82983,    0.83083,    0.83183,    0.83283,    0.83383,
0.83483,    0.83584,    0.83684,    0.83784,    0.83884,    0.83984,
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0.84685,    0.84785,    0.84885,    0.84985,    0.85085,    0.85185,
0.85285,    0.85385,    0.85485,    0.85586,    0.85686,    0.85786,
0.85886,    0.85986,    0.86086,    0.86186,    0.86286,    0.86386,
          0.86486,    0.86587,    0.86687,    0.86787,    0.86887,    0.86987,
0.87087,    0.87187,    0.87287,    0.87387,    0.87487,    0.87588,
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0.88288,    0.88388,    0.88488,    0.88589,    0.88689,    0.88789,
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0.89489,    0.8959,    0.8969,    0.8979,    0.8989,    0.8999,
0.9009,    0.9019,    0.9029,    0.9039,    0.9049,    0.90591,    0.90691,
0.90791,    0.90891,    0.90991,    0.91091,    0.91191,
          0.91291,    0.91391,    0.91491,    0.91592,    0.91692,    0.91792,
0.91892,    0.91992,    0.92092,    0.92192,    0.92292,    0.92392,
0.92492,    0.92593,    0.92693,    0.92793,    0.92893,    0.92993,
0.93093,    0.93193,    0.93293,    0.93393,    0.93493,    0.93594,
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0.94294,    0.94394,    0.94494,    0.94595,    0.94695,    0.94795,
0.94895,    0.94995,    0.95095,    0.95195,    0.95295,    0.95395,
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0.96697,    0.96797,    0.96897,    0.96997,    0.97097,    0.97197,
0.97297,    0.97397,    0.97497,    0.97598,    0.97698,    0.97798,
0.97898,    0.97998,    0.98098,    0.98198,    0.98298,    0.98398,
          0.98498,    0.98599,    0.98699,    0.98799,    0.98899,    0.98999,
0.99099,    0.99199,    0.99299,    0.99399,    0.99499,    0.996,
0.997,    0.998,    0.999,    1]), array([[ 0.94989,    0.94989,
0.94771, ...,    0,    0,    0],
[ 0.98413,    0.98413,    0.98413, ...,    0,    0,
0],
[ 0.89286,    0.89286,    0.89286, ...,    0,    0,
0]]), 'Confidence', 'Recall'])
fitness: np.float64(0.642680283890415)
keys: ['metrics/precision(B)', 'metrics/recall(B)', 'metrics/mAP50(B)',
'metrics/mAP50-95(B)']
maps: array([ 0.62862,    0.62835,    0.58933])
names: {0: 'empty_slots', 1: 'occupied_slots', 2: 'transition'}
nt_per_class: array([459, 252, 28])
nt_per_image: array([58, 58, 16])
results_dict: {'metrics/precision(B)': 0.8585568575170978, 'metrics/recall(B)':
0.902816682228447, 'metrics/mAP50(B)': 0.8878974680910833, 'metrics/mAP50-95(B)':
0.6154339300903408, 'fitness': 0.642680283890415}
save_dir: PosixPath('runs/detect/parking82_best')
speed: {'preprocess': 0.1869987930870687, 'inference': 1.9120776724023394, 'loss':
0.0003760517223593618, 'postprocess': 3.1596671896420894}
stats: {'tp': [], 'conf': [], 'pred_cls': [], 'target_cls': [], 'target_img': []}
task: 'detect'

```



```
#Project: Parking Slot Occupancy Detection
#Model: YOLOv8 (custom) - parking82_best.pt
#Video: hack17.mp4 (~2532 frames, 848x480, ~30 FPS)
```

```
#How to run:
```

```
#1) Install dependencies:
```

```
# pip install ultralytics opencv-python pandas numpy
```

```
#2) Place files in the same folder:
```

```
# - hack18.mp4
```

```
# - slots_frame1.json, slots_frame2.json, ...
```

```
#- parking82_best.pt
```

```
-# run.py
```

```
#3) Run the script:
```

```
# python run.py
```

```
#Outputs:
```

```
#- output_parking_legend_style.mp4
```

```
#- slot_status_log.csv / slot_status_log.xlsx
```

```
#Notes:
```

```
#- Frame-based JSON mapping:
```

```
# 1-655 -> slots_frame1.json
```

```
# 656-1570 -> slots_frame4.json
```

```
# 1571-2380-> slots_frame12.json
```

```
# 2381-end -> slots_frame12.json
```

```
#- Smoothing: majority vote over last 5 frames per slot
```

```
!pip install ultralytics
```



```
Collecting ultralytics
```

```
  Downloading ultralytics-8.3.189-py3-none-any.whl.metadata (37 kB)
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```
Requirement already satisfied: numpy>=1.23.0 in /usr/local/lib/python3.12/dist-package
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Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.12/dist-package
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Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.12/dist-package
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Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.12/dist-pack
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Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.12/dist-packages
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Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.12/dist-pa
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Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from
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Requirement already satisfied: polars in /usr/local/lib/python3.12/dist-packages (from
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Collecting ultralytics-thop>=2.0.0 (from ultralytics)
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1.1/1.1 MB 15.7 MB/s eta 0:00:00

Downloading ultralytics_thop-2.0.16-py3-none-any.whl (28 kB)

Installing collected packages: ultralytics-thop, ultralytics

Successfully installed ultralytics-8.3.189 ultralytics-thop-2.0.16

```
import cv2
import torch
import json
import csv
import pandas as pd
from ultralytics import YOLO
from collections import defaultdict, deque
import numpy as np
from datetime import datetime

# -----
# Parameters
# -----
INPUT_VIDEO = "hack18.mp4"
OUTPUT_VIDEO = "output_parking_legend_style.mp4"
OUTPUT_CSV = "slot_status_log.csv"
OUTPUT_XLSX = "slot_status_log.xlsx"
```

```

MODEL_PATH = "parking82_best.pt"
SMOOTHING_WINDOW = 5 # frames for majority voting

# JSON files for different frame ranges
SLOTS_FRAME1_JSON = "slots_frame1.json"
SLOTS_FRAME2_JSON = "slots_frame4.json"
SLOTS_FRAME3_JSON = "slots_frame12.json"
SLOTS_FRAME4_JSON = "slots_frame5.json"
# Frame ranges
FRAME_RANGES = [
    (1, 655, SLOTS_FRAME1_JSON),
    (656, 1570, SLOTS_FRAME2_JSON),
    (1571, 2380, SLOTS_FRAME3_JSON),
    (2381, None, SLOTS_FRAME4_JSON)
]

# -----
# Load YOLOv8 model
# -----
model = YOLO(MODEL_PATH)

# -----
# Load slot markers for all JSONs
# -----
slots_frame1 = json.load(open(SLOTS_FRAME1_JSON))
slots_frame2 = json.load(open(SLOTS_FRAME2_JSON))
slots_frame3 = json.load(open(SLOTS_FRAME3_JSON))
slots_frame4 = json.load(open(SLOTS_FRAME4_JSON))
# -----
# Video setup
# -----
cap = cv2.VideoCapture(INPUT_VIDEO)
w = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
h = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
fps = cap.get(cv2.CAP_PROP_FPS)
total_frames = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))

fourcc = cv2.VideoWriter_fourcc(*'mp4v')
out = cv2.VideoWriter(OUTPUT_VIDEO, fourcc, fps, (w, h))

# -----
# Smoothing buffers
# -----
slot_history = defaultdict(lambda: deque(maxlen=SMOOTHING_WINDOW))

# -----
# Legend config
# -----
legend_items = {
    "Empty": (0, 255, 0),
    "Occupied": (0, 0, 255),
    "Transition": (0, 255, 255)
}

```



```

        transition = (0, 255, 255)
    }

def draw_legend(frame, slot_status_counts):
    overlay = frame.copy()
    start_x, start_y = 10, 10
    width, height = 180, 100
    alpha = 0.6
    cv2.rectangle(overlay, (start_x, start_y), (start_x + width, start_y + height),
    cv2.addWeighted(overlay, alpha, frame, 1 - alpha, 0, frame)
    for idx, (label, color) in enumerate(legend_items.items()):
        y = start_y + 25 + idx * 25
        cv2.rectangle(frame, (start_x + 10, y - 12), (start_x + 30, y + 8), color,
        count = slot_status_counts.get(label, 0)
        text = f"{label}: {count}"
        cv2.putText(frame, text, (start_x + 40, y), cv2.FONT_HERSHEY_TRIPLEX, 0.5,

# -----
# Helper function to assign detection to nearest slot
# -----
def assign_slot(cx, cy, slot_markers):
    min_dist = float('inf')
    assigned_slot = None
    for slot_id, coord in slot_markers.items():
        sx, sy = coord
        dist = ((cx - sx) ** 2 + (cy - sy) ** 2) ** 0.5
        if dist < min_dist:
            min_dist = dist
            assigned_slot = slot_id
    return assigned_slot

# -----
# CSV Logging Setup
# -----
csv_file = open(OUTPUT_CSV, mode="w", newline="")
csv_writer = csv.writer(csv_file)
csv_writer.writerow(["timestamp", "frame_idx", "slot_id", "status", "confidence"])

# -----
# Process video
# -----
frame_idx = 0
while True:
    ret, frame = cap.read()
    if not ret:
        break
    frame_idx += 1
    timestamp = datetime.now().strftime("%Y-%m-%d %H:%M:%S")

    # -----
    # Select slot markers based on frame_idx
    # -----

```

```

if frame_idx <= 655:
    slot_markers = slots_frame1
elif frame_idx <= 1570:
    slot_markers = slots_frame2
elif frame_idx <= 2380:
    slot_markers = slots_frame3
else:
    slot_markers = slots_frame4

# -----
# YOLO inference
# -----
results = model(frame, verbose=False)
slot_detections = {slot_id: [] for slot_id in slot_markers.keys()}

for r in results:
    for box in r.bboxes:
        cls_id = int(box.cls[0])
        conf = float(box.conf[0])
        x1, y1, x2, y2 = map(int, box.xyxy[0])
        cx, cy = (x1 + x2) // 2, (y1 + y2) // 2

        slot_id = assign_slot(cx, cy, slot_markers)
        if slot_id is not None:
            slot_detections[slot_id].append((cls_id, conf))

# -----
# Update slot history & draw
# -----
slot_status_counts = {"Empty": 0, "Occupied": 0, "Transition": 0}

for slot_id, dets in slot_detections.items():
    if dets:
        cls_id = max(set([d[0] for d in dets]), key=[d[0] for d in dets].count)
        conf = np.mean([d[1] for d in dets])
    else:
        cls_id = 0
        conf = 1.0

    slot_history[slot_id].append(cls_id)
    final_cls = max(set(slot_history[slot_id]), key=slot_history[slot_id].count)

    # Map to label & color
    if final_cls == 0:
        label, color = "Empty", (0, 255, 0)
    elif final_cls == 1:
        label, color = "Occupied", (0, 0, 255)
    elif final_cls == 2:
        label, color = "Transition", (0, 255, 255)
    else:
        label, color = "Unknown", (255, 255, 255)

```

```

slot_status_counts[label] = slot_status_counts.get(label, 0) + 1

# Log CSV
csv_writer.writerow([timestamp, frame_idx, slot_id, label, round(conf, 2)])

# Draw square marker & label
sx, sy = slot_markers[slot_id]
size = 8
cv2.rectangle(frame, (sx - size, sy - size), (sx + size, sy + size), color,
slot_number = slot_id.replace("slot", "")
text_pos = (sx - size, sy - size - 5)
cv2.putText(frame, f"Slot {slot_number}", text_pos, cv2.FONT_HERSHEY_TRIPLEX, 1, color)

# Draw legend
draw_legend(frame, slot_status_counts)

out.write(frame)

# Optional progress print
if frame_idx % 100 == 0:
    print(f"Processing frame {frame_idx}/{total_frames}")

# -----
# Cleanup
# -----
csv_file.close()
cap.release()
out.release()

# Convert CSV → Excel
df = pd.read_csv(OUTPUT_CSV)
df.to_excel(OUTPUT_XLSX, index=False)

print(f"✅ Processed video saved as {OUTPUT_VIDEO}")
print(f"✅ Slot logs saved as {OUTPUT_CSV} and {OUTPUT_XLSX}")

🔗 Creating new Ultralytics Settings v0.0.6 file ✅
View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'
Processing frame 100/2860
Processing frame 200/2860
Processing frame 300/2860
Processing frame 400/2860
Processing frame 500/2860
Processing frame 600/2860
Processing frame 700/2860
Processing frame 800/2860
Processing frame 900/2860
Processing frame 1000/2860
Processing frame 1100/2860
Processing frame 1200/2860

```

```
Processing frame 1300/2860
Processing frame 1400/2860
Processing frame 1500/2860
Processing frame 1600/2860
Processing frame 1700/2860
Processing frame 1800/2860
Processing frame 1900/2860
Processing frame 2000/2860
Processing frame 2100/2860
Processing frame 2200/2860
Processing frame 2300/2860
Processing frame 2400/2860
Processing frame 2500/2860
Processing frame 2600/2860
Processing frame 2700/2860
Processing frame 2800/2860
```

✓ Processed video saved as output_parking_legend_style.mp4

✓ Slot logs saved as slot_status_log.csv and slot_status_log.xlsx

```
from google.colab import files
files.download("output_parking_legend_style.mp4")
```



```
from google.colab import files
files.download("slot_status_log.csv")
```



```
from google.colab import files
files.download("slot_status_log.xlsx")
```




```
// src/App.jsx
import React, { useEffect, useRef, useState, useMemo } from "react";
import VideoPlayer from "../components/VideoPlayer";
import SlotGrid2D from "../components/SlotGrid2D";
import MapView from "../components/MapView";
import { PieChart, Pie, Cell, Tooltip, Legend, RadialBarChart, RadialBar } from
"recharts";
```

```
// Pastel colors
```



```
const COLORS = { Empty: "#B5EAD7", Occupied: "#FF9AA2", Transition: "#FFD580", Total:
"#C7CEEA" };
const COLORS_RADIAL = { Empty: "#B5EAD7", Occupied: "#FFB347", Transition: "#CBA0FF" };

export default function App() {
  const videoRef = useRef(null);
  const [slotHistory, setSlotHistory] = useState([]);
  const [currentSlots, setCurrentSlots] = useState({});
  const [loading, setLoading] = useState(true);

  // Load JSON
  useEffect(() => {
    fetch("/slot_status.json")
      .then(r => r.json())
      .then(data => {
        setSlotHistory(data);
        setLoading(false);
      })
  })
}
```

```
import React from "react";
```

```
export default function SlotGrid2D({ currentFrameSlots, colors }) {  
  if (!currentFrameSlots || Object.keys(currentFrameSlots).length === 0) {  
    return <div>Loading slots...</div>;  
  }  
}
```

```
// Ensure 12 slots
```

```
const slotIds = Array.from({ length: 12 }, (_, i) => `slot${i + 1}`);
```

```
return (
```

```
  <div
```

```
    style={{
```

```
      display: "grid",
```

```
      gridTemplateColumns: "repeat(4, 60px)",
```

```
      gridGap: "10px",
```

```
      justifyContent: "center",
```

```
      alignItems: "center",
```

```
    }}  
  >
```

```
  {slotIds.map((slotId) => {
```

```
    const status = currentFramesSlots[slotId] || "Empty"; // default to Empty
```

```
    return (
```

```
      <div
```

```
        key={slotId}
```

```

// src/components/MapView.jsx
import React from "react";
import { MapContainer, TileLayer, Marker, Popup } from "react-leaflet";
import "leaflet/dist/leaflet.css";
import L from "leaflet";

// custom marker
const markerIcon = new L.Icon({
  iconUrl: "https://unpkg.com/leaflet@1.7.1/dist/images/marker-icon.png",
  iconSize: [25, 41],
  iconAnchor: [12, 41],
  popupAnchor: [1, -34],
});

export default function MapView() {
  // Coordinates of Saranathan College of Engineering 4-wheeler parking
  const position = [10.7924, 78.6821]; // adjust if you have more precise GPS

  return (
    <div style={{ width: "500px", height: "300px", margin: "0 auto", borderRadius: 12, overflow: "hidden", boxShadow: "0 2px 8px
    rgba(0,0,0,0.25)" }}>
      <MapContainer
        center={position}
        zoom={19} // close-up zoom
        style={{ height: "100%", width: "100%" }}
      >
        { /* Satellite view */ }
        <TileLayer
          url="https://server.arcgisonline.com/ArcGIS/rest/services/World_Imagery/MapServer/tile/{z}/{y}/{x}"
          attribution="© Esri &mdash; Source: Esri, Maxar, Earthstar Geographics, USDA, USGS, AeroGRID"
        />
        <Marker position={position} icon={markerIcon}>
          <Popup>4-Wheeler Parking - Saranathan College of Engineering</Popup>
        </Marker>
      </MapContainer>
    </div>
  );
}

```

```
// src/components/VideoPlayer.jsx
import React from "react";

export default function VideoPlayer({ videoRef, videoSrc, onTimeUpdate }) {
  return (
    <video
      ref={videoRef}
      src={videoSrc}
      controls
      width={480}
      height={270}
      style={{ borderRadius: 12, background: "#000" }}
      onTimeUpdate={onTimeUpdate}
    />
  );
}
```

```

// Charts.jsx
import React from "react";
import { PieChart, Pie, Cell, Tooltip, ResponsiveContainer } from "recharts";

const COLORS = { Empty: "#90EE90", Occupied: "#FF6B6B", Transition: "#FFD580" };

export default function Charts({ latestData, currentFrameSlots }) {
  // Prepare pie chart from currentFrameSlots
  const counts = { Empty: 0, Occupied: 0, Transition: 0 };
  Object.values(currentFrameSlots).forEach(s => counts[s] ? counts[s]++ : null);

  const pieData = [
    { name: "Empty", value: counts.Empty },
    { name: "Occupied", value: counts.Occupied },
    { name: "Transition", value: counts.Transition },
  ];

  return (
    <div style={{ width: 300, height: 300 }}>
      <h3 style={{ textAlign: "center" }}>Slot Status Pie Chart</h3>
      <ResponsiveContainer>
        <PieChart>
          <Pie data={pieData} dataKey="value" cx="50%" cy="50%" outerRadius={100} label>
            {pieData.map((entry, index) => (
              <Cell key={index} fill={COLORS[entry.name]} />
            ))}
          </Pie>
          <Tooltip />
        </PieChart>
      </ResponsiveContainer>
    </div>
  );
}

```